Venkata Lakshmi Parimala Pasupuleti

Agentic AI Engineer · Mortgage & Real Estate Domain · Multi-Step Agents · Eval Pipelines · LangGraph · Python · AWS

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Agentic AI engineer with **8 years in real estate and lender-facing SaaS** (Entrata, 100K+ MAU), now shipping **production agentic AI systems at Intuit**. Built LeaseI: an **8-agent LangGraph platform** for real estate document processing — multi-step agents, **eval pipeline infrastructure**, HITL escalation, idempotency/retry reliability, and full observability — achieving **93.6% precision across 1,240 lease documents**. At Entrata, built **lender-facing HUD and Tax Credit certification workflows** (income verification, condition clearing, document collection) across 1.6M+ property records — the direct operational analog to Zillow's mortgage fulfillment pipeline. AgentProbe is an open-source **LLM eval toolkit** with CI regression harness and LLM-as-judge scoring. At Intuit, builds **eval harnesses with golden datasets**, production monitoring for model drift, and agentic AI features (Anthropic Claude + Python/FastAPI + AWS). **Strong infrastructure instincts**: RabbitMQ async at 99.5% uptime, idempotency/retries, structured logging, observability. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Agentic AI & Evals: Multi-Step Agentic Workflows, LangGraph, LangChain, LLM Orchestration, Eval Pipeline Design, LLM-as-Judge, Golden Datasets, CI Regression Testing, HITL Escalation, Guardrails, Structured Outputs, Observability, Prompt Engineering, Anthropic Claude, OpenAI, Groq, Ollama

Backend & Infrastructure: Python (9+ yrs), FastAPI, Flask, Java, Spring Boot, Node.js, REST APIs, RabbitMQ (pub/sub, idempotency, retries), Async Processing, Docker, Kubernetes, AWS (S3, EC2, RDS, Lambda, SQS, API Gateway, IAM), Terraform, GitHub Actions CI/CD, Prometheus, Grafana, OpenTelemetry, PostgreSQL, Snowflake, Redis, SQLite

Real Estate & Lender Domain: HUD/Tax Credit Certification Workflows, Income Verification, Condition Clearing, Document Collection, Third-Party Lender Integrations, LIHTC/HCV/HUD Regulatory Requirements, PHI Compliance & RBAC, 1.6M+ Financial Records

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — Agentic AI · Eval Infrastructure · Financial Analytics

- Build **production agentic AI systems** with Anthropic Claude (Python/FastAPI): financial data summaries, natural-language queries; implement **guardrails** (input validation, sensitive-data handling, graceful fallbacks on model output failure) and **trace-level observability** via OpenTelemetry and Prometheus; deploy via GitHub Actions CI/CD to AWS.
- Build and run **LLM-as-judge eval harnesses** (offline evals + golden datasets + CI regression) and **production monitoring** to catch hallucination and model drift before every release; **define what correct looks like, build tooling to check it, use that signal to improve prompts and models**; build end-to-end ML pipelines (scikit-learn, XGBoost) on Snowflake through prediction APIs.

Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

Senior Software Engineer — Lender-Facing Workflows · Document Integrations · Real Estate Data

- Built **lender-facing HUD and Tax Credit certification systems**: income verification, document collection, condition clearing, and regulatory certification workflows across 1.6M+ property records — direct analog to Zillow Mortgage Fulfillment pipeline. Served as **primary technical contact for federal data source integrations**: scoped data contracts with HUD/Tax Credit teams, debugged third-party API issues, coordinated go-lives, and resolved production incidents end-to-end.
- Built **RabbitMQ async processing** (pub/sub, idempotency, retry, dead-letter) at **99.5% uptime** across thousands of tenant records; **optimized 200+ SQL queries (30% gain)**; Docker microservices; AWS S3 migration with Libsodium RBAC for PHI — **100% compliance, zero incidents**; reviewed 300+ PRs, mentored junior engineers; **120+ Agile sprints**.

PROJECTS

LeaseI — 8-Agent Agentic AI Platform with Eval Infrastructure

Multi-Step Agents · Eval Pipeline · HITL

Python · LangGraph (8-agent) · FastAPI · Hybrid RAG (TF-IDF + semantic) · Prompt Caching · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Multi-step agentic workflow** (triage → retrieval → reasoning → validation → HITL escalation) over 1,240 real estate documents; **eval pipeline infrastructure** with LLM-as-judge CI harness and labeled test datasets — **93.6% precision, 87.9% F1**; production reliability via **idempotency, retries, graceful degradation**; structured logging and full observability; guardrails routing regulated decisions to humans; every answer traced to source clause.

AgentProbe — LLM Eval Toolkit with CI Regression Harness

Eval Infrastructure · Model Routing

Python · FastAPI · LangGraph · ChromaDB · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; **CI golden-dataset regression harness** measuring every model and prompt change before ship; **LLM-as-judge scoring** (accuracy, hallucination rate, retrieval quality, completeness, compliance); dynamic model routing across 5 LLM providers; 3 LangGraph agents. Designed to **define what correct looks like and build tooling to verify it**.

AI on FHIR — NLP-to-API with Third-Party Integration

API Integration · Auth · Full-Stack

Python · Flask · React · TypeScript · Docker · BERT · LLMs · FHIR REST APIs — github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates EN/ES natural-language queries into **live third-party API calls** via LLMs + BERT; **HIPAA-style RBAC with fine-grained authorization**; Docker containerized with reliable error handling and fallbacks; cut manual lookup time ~60%. Demonstrates **reliable integration with external platforms within their constraints**.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · ML pipelines, LLM evaluation, agentic systems, statistical modeling, distributed systems

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017